

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech (EC) Examination****May 2017****EC 407 Radar Systems (Elective Subject)****Date: 27.05.2017, Saturday****Time: 01:30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the question below. [07]**

- a. Define following terms with reference to Radar. [05]
- 1) Probability of false alarm
 - 2) Probability of Miss
 - 3) 'A' type Display
 - 4) Radio Direction Finder
 - 5) Polarization error
- b. In search Radar _____ antenna can be used. [01]
- | | |
|-----------------|-------------------------------|
| (a) Wide beam | (b) Wide and narrow beam both |
| (c) Narrow beam | (d) None of these |
- c. Ku band having a _____ frequency range. [01]
- | | |
|---------------|--------------|
| (a) 4-8 GHz | (b) 8-12 GHz |
| (c) 12-18 GHz | (d) 8-13 GHz |

Q – 2.a A circular loop with 10 turns at the height of 1m. Find the operating frequency if the frequency if the diameter of loop is 2 meters. **[04]****Q – 2.b Answer any two. [10]**

- (i) What is receiver noise? Derive the Radar range equation in terms of bandwidth.
- (ii) Draw and explain block diagram of conventional pulse radar with a super-heterodyne receiver.
- (iii) An MTI radar is operating with 0.1Ghz with a PRF of 200 Hz. Calculate the lower three blind speeds.

Q - 3 Answer any two. [14]

- a. Consider Radar with pulse repetition frequency 1000 Hz. (a) Find the time duration between two pulses. (b) Suppose an echo from a distant object is received 20 μ sec after a pulse is transmitted, what is the distance of the object from the radar? (c) Is there a second -time-around echo from this object?
- b. Draw block diagram of Range Gated Doppler filters and explain it in detail.

- c. Why we require I & Q Channel as a separate channel for digital signal processing?
Explain it with suitable waveforms.

SECTION – II

Q - 4 Answer the question below. [07]

- a. What is Hyperbolic Navigation? [01]
- b. Write any Three names of hyperbolic Navigation system. [01]
- c. What is the frequency range of LORAN-C system? [01]
- d. Write a full name of DECCA. [01]
- e. What is the use of Decometer in Radar? [01]
- f. What is 'Lane' in Omega system? [02]

Q – 5.a Draw and explain block diagram of altimeter. [05]

OR

Q – 5.a Explain the need and basic principle of FMCW Radar. [05]

Q – 5.b Derive the equation of Beat frequency from appropriate waveforms. [04]

Q – 5.c What is Doppler effect? Explain function of each block in detail for Pulse Doppler Radar. [05]

OR

Q – 5.c Describe different types of constraints in operation of FMCW Radar. [05]

Q – 6.a Explain in brief the Airport Radar. [05]

Q – 6.b Write a short note on Meteorological Radar. [05]

Q – 6.c Explain in brief SLAR (Side Looking Airborne Radar). [04]

OR

Q – 6.a Write a short note on Airborne Radar. [05]

Q – 6.b Explain in brief the Remote Sensing Radar. [05]

Q – 6.c Derive the Doppler Radar Range Equation. [04]
